

DexSafeDagger: Safer Policy Distillation from RGB Observations Using Q-Value Failure Prediction–Guided Teacher Intervention

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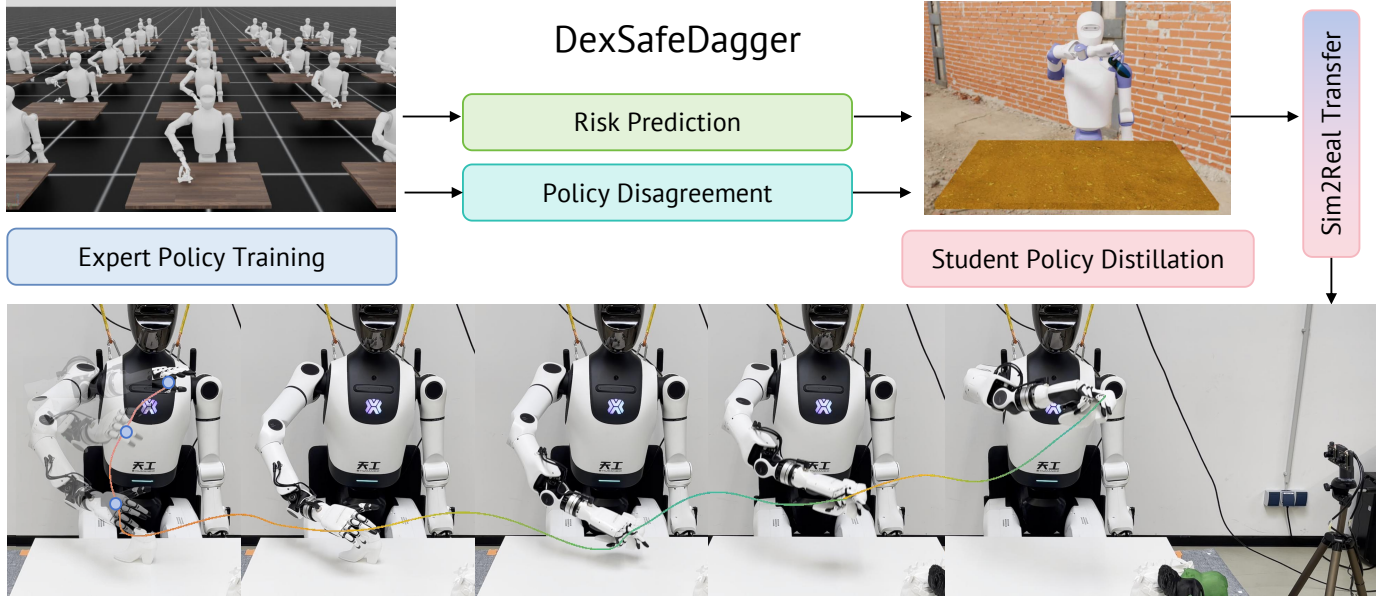


Fig. 1: **DexSafeDagger overview.** We propose a teacher-intervened online distillation pipeline that condenses teacher skills into a deployable RGB-based student policy. The unsafe exploration rate is substantially suppressed in the early training phase via a customized intervention mechanism that combines teacher–student action disagreement with a learned risk predictor.

Abstract—Learning dexterous manipulation is challenging due to high-dimensional action spaces and complex contact dynamics. In addition, safety-unaware exploration during learning can lead to hardware damage, unstable interactions, and hazardous contact events, posing a significant barrier to real-world deployment. To address this challenge, we propose a safety-aware online policy distillation framework that transfers reinforcement learning (RL) teacher policies to a vision-based student policy. Our approach introduces a teacher-intervened mixed rollout mechanism that integrates teacher–student action disagreement with a learned Q-value-based risk predictor to detect high-risk states and proactively trigger teacher takeover. This design enables safer exploration while preserving the benefits of online interactive imitation learning. Experiments on grasp-and-lift tasks show that our method significantly improves training safety, reducing unsafe episodes by up to 80% during the early stages of training compared with vanilla DAgger-based distillation. Meanwhile, the distilled student policy achieves comparable lift success rates and demonstrates robustness suitable for real-world robotic deployment.

I. INTRODUCTION

Dexterous robotic hands typically possess a large number of degrees of freedom (DoFs) and exhibit complex contact

dynamics, enabling them to perform fine-grained and diverse manipulation behaviors, such as multi-finger grasping [1], in-hand reorientation [2, 3], and precise control [4]. However, the high-dimensional action space and intricate interaction dynamics substantially increase the difficulty of control and policy learning. In recent years, reinforcement learning (RL)-based approaches [5, 3, 6, 7] have achieved remarkable progress in dexterous grasping and manipulation tasks. Leveraging large-scale parallel simulation and high-fidelity physics engines, RL enables the automatic learning of complex control policies in simulated environments without the need for manually designed controllers. These methods have demonstrated strong generalization capabilities across multi-object and multi-task scenarios.

Despite these advances, existing reinforcement learning frameworks primarily optimize for task success rates and rarely incorporate explicit modeling or constraints related to safety-critical behaviors. During training, common failure modes include severe collisions between the hand and the environment, unintended object ejection due to excessive contact forces, and loss of stable hand posture [8, 9]. Such unsafe behaviors not only reduce training efficiency but also introduce potential risks during real-world deployment. Therefore, how to reduce unsafe exploration while maintaining learning efficiency remains a

critical challenge in dexterous manipulation.

To mitigate the performance gap between simulation and real-world deployment, policy distillation has become a widely adopted paradigm in recent years. The core idea is to first train a high-performing expert policy in simulation using privileged state information, and then distill its decision-making behavior into a student policy that relies solely on visual observations or partially observable inputs. This process enables modality transfer from privileged state information to perception-based inputs, facilitating efficient real-world deployment. Policy distillation reduces reliance on accurate state estimation and allows knowledge integration from multiple teacher policies, thereby substantially improving execution stability and generalization performance in real-world environments. Consequently, it has gained increasing attention in dexterous grasping and complex manipulation tasks. However, existing distillation methods [10, 11] typically rely on behavior cloning or the standard DAgger framework for data aggregation. Their primary objective is to mitigate performance degradation caused by distribution shift, rather than explicitly enforcing safety constraints during student policy learning. During distillation, the student policy may deviate from the teacher’s state–action distribution and enter regions that are either insufficiently covered by the teacher or inherently high-risk, leading to unsafe behaviors such as severe collisions or unstable hand configurations. Therefore, introducing explicit safety mechanisms into the distillation process—so that high-risk exploration can be avoided without sacrificing learning efficiency—remains an important and largely underexplored problem. Existing safety-aware distillation approaches, such as SafeDagger [12], have primarily focused on autonomous driving scenarios and have not been systematically studied in the context of dexterous manipulation.

The contributions of this work are summarized as follows:

- We propose a safety-sensitive framework for distilling RL-acquired teacher policies into a single RGB-based student policy. To the best of our knowledge, this is the first attempt in dexterous manipulation to incorporate a mixed intervention mechanism that combines teacher–student action disagreement with a Q-value-based risk predictor for proactive teacher takeover in high-risk regions.
- We propose a safety-aware RL reward design that explicitly accounts for risky events in arm–hand control scenarios, including harmful collisions, object escape, palm flipping, and the hand moving too far from the workspace, thereby encouraging safer exploration during teacher training.
- We empirically demonstrate that combining disagreement and risk prediction substantially reduces unsafe exploration compared with vanilla DAgger, achieving a relative reduction of **80%** in unsafe exploration rate in the early training phase. We also observe a **5%** improvement in unsafe exploration rate compared with disagreement-only SafeDagger.

II. RELATED WORK

A. Policy distillation

Policy distillation transfers behavior from a high-capacity teacher to a compact student and has been adopted from vision to robotics. In robotics, the teacher can leverage privileged state and simulator access during training, while the student is distilled to operate from on-board observations (e.g., RGB). DexTrAH-RGB [11] distills a privileged RL teacher into an RGB-based policy via DAgger-based imitation learning, while UniGraspTransformer [13] distills offline trajectories from object-specific teachers into a unified Transformer policy.

B. Interactive imitation learning

Many imitation learning methods operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution [14, 15]. In sequential decision-making, however, distributional shift induces covariate shift and compounding errors. Interactive imitation learning [16] mitigates this issue by collecting supervision online under the learner-induced state distribution.

DAgger [10] addresses covariate shift by iteratively rolling out the learner and relabeling visited states with an expert, thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over in high-risk states to reduce unsafe exploration. Recent variants [17, 18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision. Fully autonomous expert relabeling during online distillation remains rare in robotic manipulation, especially for dexterous, contact-rich tasks.

III. PROBLEM STATEMENT AND METHODS

A. Problem Formulation

We consider online interactive imitation learning with teacher intervention in a partially observable Markov decision process (POMDP) $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, \Omega, r, \gamma)$, with privileged teacher state $s_t \in \mathcal{S}$, deployable student observation $o_t \in \mathcal{O}$ (RGB), action $a_t \in \mathcal{A}$, observation model $o_t \sim \Omega(\cdot | s_t)$, and dynamics $s_{t+1} \sim P(\cdot | s_t, a_t)$. During training, we assume access to a set of privileged teacher policies $\{\pi_{T_i}\}_{i=1}^K$, whereas at deployment the student policy π_S must act solely from o_t . Our goal is to distill the collective teacher behavior into an RGB-based student that achieves high task success while minimizing unsafe exploration during training.

$$\begin{aligned} \min_{\substack{\pi_S \in \Pi_S \\ \mathcal{G} \in \mathcal{G}}} & \left[\underbrace{-\mathbb{E}_{\tau \sim p_{\pi_G}} [\mathbb{I}_{\text{succ}}(\tau)]}_{\text{maximize success}} + \lambda \underbrace{\mathbb{E}_{\tau \sim p_{\pi_G}} \left[\sum_{t=0}^{T-1} \mathbb{I}_{\text{unsafe}}(s_t, a_t) \right]}_{\text{minimize unsafe}} \right] \\ \text{s.t. } & a_t = g_t a_t^T + (1 - g_t) a_t^S, \\ & g_t = \mathcal{G}(s_t, o_t) \in \{0, 1\}, \\ & s_{t+1} \sim P(\cdot | s_t, a_t), \quad o_t \sim \Omega(\cdot | s_t). \end{aligned} \tag{1}$$

where $g_t = 1$ indicates teacher takeover and $g_t = 0$ denotes student control. The state distribution p_{π_G} is induced by the

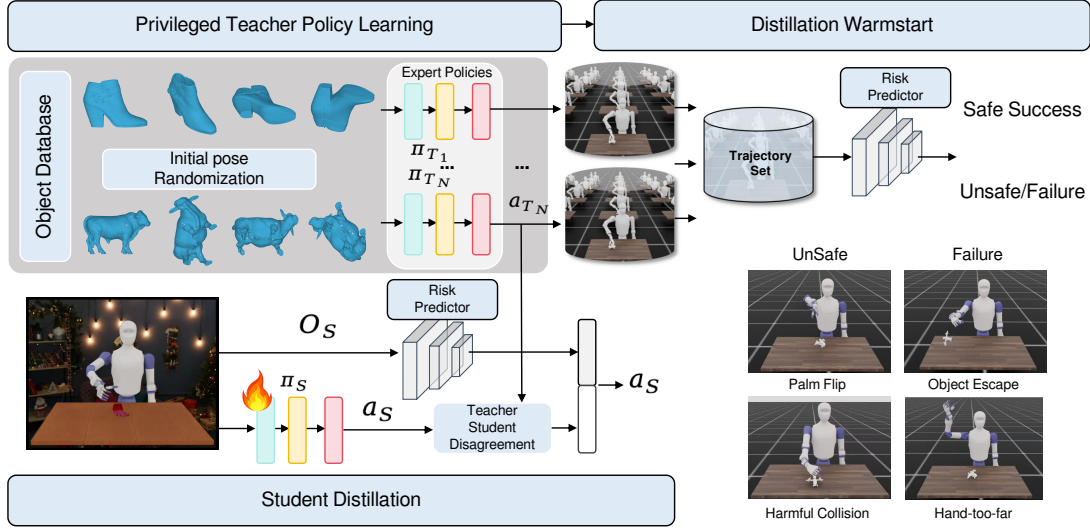


Fig. 2: Overview of the proposed teacher-student learning pipeline. A privileged reinforcement learning teacher is trained in simulation and subsequently distilled into an RGB-based visuomotor policy for real-world deployment.

gated rollout policy $\pi_{\mathcal{G}}$, which switches between the teacher and student policies according to the intervention gate $g_t = \mathcal{G}(s_t, o_t)$.

We formalize task success via a trajectory-level predicate $\mathbb{I}_{\text{succ}}(\tau) \in \{0, 1\}$, which is 1 if the object is grasped, lifted, and held above a height threshold for a consecutive duration. Safety is governed by early termination conditions: letting $\mathbb{I}_{\text{unsafe}}(s_t, a_t) \in \{0, 1\}$ indicate whether any unsafe event (e.g., harmful collision, object escape, palm flip, hand-too-far) occurs at time t , the trajectory is safe if $\prod_{t=0}^{T-1} (1 - \mathbb{I}_{\text{unsafe}}(s_t)) = 1$.

B. Overview of the Distillation Pipeline

Fig. 2 summarizes our teacher-student distillation pipeline, which consists of three components: (i) privileged teacher policy training in simulation (Sec. C), (ii) risk predictor learning to estimate lift success under student actions (Sec. D), and (iii) online student distillation with teacher intervention (Secs. E–F). We warm-start the risk predictor using an offline dataset collected under full teacher control and continue updating it periodically during online distillation using mini-batches sampled from a ring replay buffer. During interactive rollouts, an intervention gate $\mathcal{G}$ switches control between teacher and student based on an OR-rule combining a learned risk estimate and teacher-student action disagreement. The student is then optimized online using per-timestep teacher labels collected under the gated rollout policy $\pi_{\mathcal{G}}$. Pseudo-code is provided in Alg. 1.

C. Privileged Teacher Policy Training

Teacher policy learning is formulated as a reinforcement learning problem with transition tuples $\langle s_t, a_t, r_t, s_{t+1} \rangle$, where s_t is the privileged state, a_t is the action, r_t is the reward, and s_{t+1} is the next state. The objective is to maximize the

Algorithm 1 Safe Policy Distillation with Risk-Guided Teacher Intervention

- 1: Train teacher policy π_T via reinforcement learning
- 2: Collect offline dataset $\mathcal{D}_0$ using π_T
- 3: Pretrain risk predictor $\hat{Q}(s, a)$ on $\mathcal{D}_0$
- 4: **for** iteration $k = 1 \dots N$ **do**
- 5: Observe student observation o_t
- 6: Compute student action $a_t^S \sim \pi_S(\cdot | o_t)$
- 7: Query teacher action $a_t^T \sim \pi_T(\cdot | s_t)$
- 8: Compute disagreement $\delta_t = \|a_t^S - a_t^T\|_2$
- 9: Evaluate risk score $\hat{Q}(s_t, a_t^S)$
- 10: **if** $d_t > \tau_D$ **or** $\hat{Q}(s_t, a_t^S) < \tau_{\text{success}}$ **then**
- 11: Execute teacher action $a_t = a_t^T$
- 12: **else**
- 13: Execute student action $a_t = a_t^S$
- 14: **end if**
- 15: Store (o_t, a_t^T) in ring replay buffer $\mathcal{D}$
- 16: Update student policy π_S via imitation loss
- 17: Update risk predictor $\hat{Q}$ using replay buffer
- 18: **end for**

expected discounted return,

$$\pi_{\theta}^* = \arg \max_{\pi_{\theta}} \mathbb{E}_{\pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \quad (2)$$

We use a shaped reward $r_t = \sum_{k=1}^{K_r} r_{t,k}$ to encourage stable grasp acquisition, object lifting, and goal reaching (Table I).

The teacher is trained in simulation using an asymmetric actor-critic setup (Table II): the actor receives noisy and partial proprioceptive observations that approximate realistic sensing, while the critic is provided with privileged full state of both robot and object, available only in simulation.

Episodes terminate at

$$T = \min \{t \mid \mathbb{I}_{\text{unsafe}}(s_t) = 1 \text{ or } t \geq T_{\text{max}}\}, \quad (3)$$

where $T_{\text{max}} = 10$ s is the timeout horizon. Early termination

TABLE I: Reward components used for teacher policy training.

Term	Name	Purpose
r_{dist}	Hand distance	Encourages the hand to approach the object.
r_{curl}	Finger curl	Biases fingers toward a grasp-ready configuration.
r_{align}	Palm alignment	Aligns palm orientation with the palm-down direction.
r_{contact}	Contact bonus	Rewards stable contact involving thumb and fingers.
r_{near}	Stay-close	Encourages the hand to remain close after reaching the object.
r_{goal}	Goal tracking	Encourages moving the grasped object toward the goal pose.
r_{lift}	Lift	Rewards vertical lifting after successful contact.
r_{action}	Action smoothness	Penalizes rapid action changes for stable control.

conditions include palm flip, harmful collision, object escape, and hand-too-far (see Sec. A), and are treated as unsafe events for both teacher training and subsequent intervention decisions.

D. Risk Predictor

Action disagreement alone cannot anticipate downstream failures, particularly when the student action is locally similar to the teacher action yet triggers unstable contact transitions. To address this limitation, we learn a risk predictor that conservatively estimates the probability of *lift success*. The predictor is used only during training to guide teacher intervention and is not required at deployment time.

The risk predictor is designed to estimate risk from multimodal inputs. It takes as input a state–action pair (s_t, a_t) , where s_t comprises robot proprioceptive features concatenated with image embeddings produced by a visual encoder to provide semantic information. The visual encoder is trained end-to-end during distillation and regularized via an auxiliary object-position prediction loss (Sec. F). The action a_t is the *executed* action at time t (teacher or student, depending on the intervention gate). Following a clipped double-critic design, we maintain two critics Q_1 and Q_2 implemented as ReLU MLPs with hidden sizes [256, 128], each outputting a scalar logit $z_i(s, a) \in \mathbb{R}$ mapped to a success probability via a temperature-calibrated sigmoid, $Q_i(s, a) = \sigma(z_i(s, a)/T)$ for $i \in \{1, 2\}$, where $T = 4$ is an empirically set calibration temperature. We then compute

$$\hat{Q}(s, a) = \min(Q_1(s, a), Q_2(s, a)). \quad (4)$$

which provides a conservative estimate of lift success (lower values indicate higher risk).

The predictor is pretrained in a warm-start phase using an offline dataset of transitions $(s_t, a_t, s_{t+1}, a_{t+1}, u_t, d_t, h_t)$. Here, (s_t, a_t) and (s_{t+1}, a_{t+1}) provide the current and next state–action pairs required by SARSA-style bootstrapping, $d_t \in \{0, 1\}$ is the episode termination flag, and $h_t \in \{0, 1\}$ indicates whether the next action a_{t+1} is defined (valid) for bootstrapping. The label $u_t \in \{0, 1\}$ encodes *safe lift-event* attainment by time t : $u_t = 1$ if the object has been lifted above a height threshold for a consecutive duration (as in $\mathbb{I}_{\text{succ}}$) and no unsafe event has occurred up to time t ; otherwise $u_t = 0$.

During online distillation, newly collected transitions are appended to a ring buffer (sized $100k$) at every environment step. The predictor critics are updated periodically: every 500 environment steps, we perform 20 gradient updates using mini-batches sampled from the buffer using Adam with learning rate 10^{-3} .

Training uses SARSA-style bootstrapped regression with an effective termination signal $d_t^{\text{eff}} = \max(d_t, 1 - h_t)$ and one-step target

$$y_t = \text{clip}\left(u_t + \gamma(1 - d_t^{\text{eff}}) \min_{i \in \{1, 2\}} Q_i^{\text{targ}}(s_{t+1}, a_{t+1}), 0, 1\right), \quad (5)$$

where Q_i^{targ} are slowly-updated target networks (Polyak averaging) used only to stabilize bootstrapping, and the min operator mitigates overestimation. The critics are optimized with MSE $\mathcal{L}_{Q_i} = \mathbb{E}[w_t(Q_i(s_t, a_t) - y_t)^2]$, with total loss $\mathcal{L} = \mathcal{L}_{Q_1} + \mathcal{L}_{Q_2}$ and target updates $\theta' \leftarrow \rho\theta' + (1 - \rho)\theta$ ($\rho = 0.995$).

During interactive distillation, the predictor estimates the task success probability of executing the *student* action: given s_t and $a_t^S \sim \pi_S(\cdot | o_t)$, we compute $\hat{Q}(s_t, a_t^S)$ and trigger teacher takeover when it falls below a threshold (Sec. E).

E. Intervention Mechanism

At each timestep, we compute (i) a risk score using the learned predictor and (ii) a teacher–student disagreement score. Disagreement is defined as $\delta_t := \|a_t^S - a_t^T\|_2$ where $a_t^T \sim \pi_T(\cdot | s_t)$ and $a_t^S \sim \pi_S(\cdot | o_t)$ are the teacher and student action outputs. The risk criterion flags states whose predicted success under the student action is low, i.e., $\hat{Q}(s_t, a_t^S) < \tau_{\text{success}}$.

We instantiate the intervention gate $\mathcal{G}$ as an OR-rule that triggers teacher takeover when either the predictor indicates high risk or the student deviates substantially from the teacher:

$$g_t = \mathbb{I}[\hat{Q}(s_t, a_t^S) < \tau_{\text{success}} \vee d_t > \tau_D]. \quad (6)$$

Intuitively, disagreement detects distribution shift, while the risk predictor anticipates downstream failures even when the instantaneous action discrepancy is small.

We set τ_D to the 90th percentile of the teacher–student disagreement δ_t measured on the warm-start dataset. We set τ_{success} to a low percentile (e.g., the 10th) of $\hat{Q}(s_t, a_t^T)$ computed over successful warm-start rollouts. Both thresholds are kept fixed throughout training. These percentiles are chosen empirically to balance intervention frequency and exploration efficiency.

F. Student Policy Distillation

Student learning is posed as supervised imitation from teacher-labeled transitions collected under the gated rollout policy $\pi_{\mathcal{G}}$. The student observes RGB images (two stereo views) together with robot proprioception.

We optimize a combined objective $\mathcal{L} = \mathcal{L}_{\text{action}} + \mathcal{L}_{\text{aux}}$, where the auxiliary loss predicts object position to regularize the visual encoder, $\mathcal{L}_{\text{aux}} = \|\hat{\mathbf{x}}_{\text{obj}} - \mathbf{x}_{\text{obj}}\|_2$.

For action imitation, we use a weighted L_2 loss, $\mathcal{L}_{\text{action}} = \|\mathbf{a}_S - \mathbf{a}_T\|_W^2 = (\mathbf{a}_S - \mathbf{a}_T)^\top W (\mathbf{a}_S - \mathbf{a}_T)$, with $W = \text{diag}(w_1, \dots, w_d)$. We set $w_j = \sigma_{0,j}^{-2}$ using a reference scale $\sigma_{0,j}$ (e.g., the teacher action standard deviation on dimension j), yielding $\mathcal{L}_{\text{action}} = \sum_{j=1}^d (a_{S,j} - a_{T,j})^2 / \sigma_{0,j}^2$.

Training proceeds in two stages. (i) *Warm-start*: we collect an offline dataset (e.g., 100k transitions) under full teacher control ($g_t \equiv 1$) to pretrain the risk predictor. (ii) *Online distillation*: we perform interactive rollouts under π_G and query teacher labels on intervened steps. The student is updated online using the current-step teacher label only; we do *not* maintain a replay buffer or DAgger-style aggregated dataset for student updating. The design reduces the effect of stale supervision and stabilizes online imitation updates.

IV. EXPERIMENT

A. Evaluation Metrics

We designed the following metrics to evaluate both teacher and student policy performance. Let $\tau \in \{1, \dots, N\}$ index evaluation episodes and $t \in \{0, \dots, T_\tau\}$ index timesteps. Unless otherwise stated, we run the evaluation episodes with a horizon of 10 s.

a) Lift success (ever-lifted with hold).

Let h_t^τ be the object height above the table and H a lift threshold. An episode is considered successful if the object remains above H for at least N_{hold} consecutive steps. We define $\mathbb{I}_{\text{lift}}^\tau = \mathbb{I}[\exists t : \forall k \in \{0, \dots, N_{\text{hold}} - 1\}, h_{t+k}^\tau \geq H]$ and report: $\text{LiftSucc} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{lift}}^\tau$.

b) Unsafe episode rate.

As stated above, an episode may terminate early due to one of the following conditions: palm flipped ($\mathbb{I}_{\text{flip}}^\tau$), harmful collision ($\mathbb{I}_{\text{coll}}^\tau$), object escape ($\mathbb{I}_{\text{obj}}^\tau$), or hand too far ($\mathbb{I}_{\text{hf}}^\tau$). Let $\mathcal{M} = \{\text{coll}, \text{obj}, \text{flip}, \text{hf}\}$ denote the set of termination modes, and let $\mathbb{I}_m^\tau \in \{0, 1\}$ indicate whether mode $m \in \mathcal{M}$ occurs in episode τ . An episode is unsafe if any termination mode is triggered: $\mathbb{I}_{\text{unsafe}}^\tau = \mathbb{I}[\bigvee_{m \in \mathcal{M}} \mathbb{I}_m^\tau = 1]$. We report: $\text{UnsafeRateAvg} = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_{\text{unsafe}}^\tau$.

c) Failure-mode decomposition.

Each episode is assigned a single terminal mode $m^\tau \in \mathcal{M}$ from the set of unsafe events (collision, object escape, palm flip, hand-too-far) as well as timeout/success where applicable. We define $\mathbb{I}_m^\tau = \mathbb{I}[m^\tau = m]$ and report $\text{FailureModeRate}(m) = \frac{1}{N} \sum_{\tau=1}^N \mathbb{I}_m^\tau, m \in \mathcal{M}$.

When multiple unsafe conditions are triggered in the same timestep, we record the first triggered termination condition (using a fixed priority order) as the episode’s failure mode.

B. Simulation Experiment Setup

Simulation experiments use single-arm control on a Tian-gong 2 Pro humanoid equipped with InspireHand RH56DfX; the action space is 13-D (7 arm DoF + 6 hand DoF). The RGB student additionally receives two 320×240 camera streams at 30 fps, encoded by a Siamese ResNet-18 visual backbone with view fusion into a 128-D feature appended to the actor observation. Teacher training and student distillation

TABLE II: Observation spaces used for teacher RL training under the asymmetric actor–critic framework.

Observation Component	Actor	Critic
Robot DOF positions	13 (noisy)	13
Robot DOF velocities	13 (noisy)	13
Hand link positions	18	18
Hand link velocities	18	36 (twist)
Hand contact forces	–	3
Measured joint torque	–	19
Object position	3	3
Object rotation (quaternion)	4	4
Object velocity	–	6 (twist)
Goal position	3	3
Object scale	1	1
Previous actions	13	13
Total dimension	86	132

TABLE III: Teacher policy performance across various objects (tested for 200 episodes)

Percentage %	WindCart	Boot	ToolBox	Cow
Lift Success	93.5	100.0	88.5	87.0
Unsafe Rate Avg	6.5	13.5	13.5	13.5
Harmful collision	0.0	0.0	6.5	0.0
Object escape	6.5	6.5	6.5	13.5
Palm flipped	0.0	6.5	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0

are conducted in Isaac Sim 5.0.0 / Isaac Lab 2.2.1 on a single RTX 5090 GPU.

We evaluate 10 objects from the Omniverse Asset Library. Each teacher is trained for 12,000 episodes (3.5 h), achieving 72–100% lift success and 5–30% unsafe rate (Tab. 3). During distillation, we randomize lighting intensity, direction and background textures as well as object material and table appearance (DexTraH-RGB-style domain randomization); the distillation stage runs 100,000 episodes in 2.5 h.

C. Ablation Studies

The following ablation studies are conducted to justify our pipeline design. We distill the teacher policies trained during the RL phase to obtain a compact and generalizable student policy. The RL training baseline follows the DexTraH-RGB pipeline, and we compare it against our customized variant. We evaluate three distillation variants—Vanilla DAgger, Vanilla SafeDAgger, and our proposed pipeline—in terms of lift-success growth, unsafe-episode rate, and intervention rate (defined as the fraction of timesteps in which teacher takeover occurs, i.e., $\beta = \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{I}[g_t = 1]$, averaged over evaluation episodes) during training. Since unsafe-episode rate is our primary metric, we report not only the average rate across all test objects but also a per-object, failure-mode decomposition to assess the suitability of our pipeline under different failure conditions. We first define the ablation setups below:

(a) DexTraH-RGB RL teacher + Vanilla DAgger. The teacher is trained using the DexTraH-RGB RL recipe, and the student is learned via Vanilla DAgger. This is the original DexTraH-RGB pipeline used for the KUKA–Allegro setup. During data collection, the student fully controls the rollout

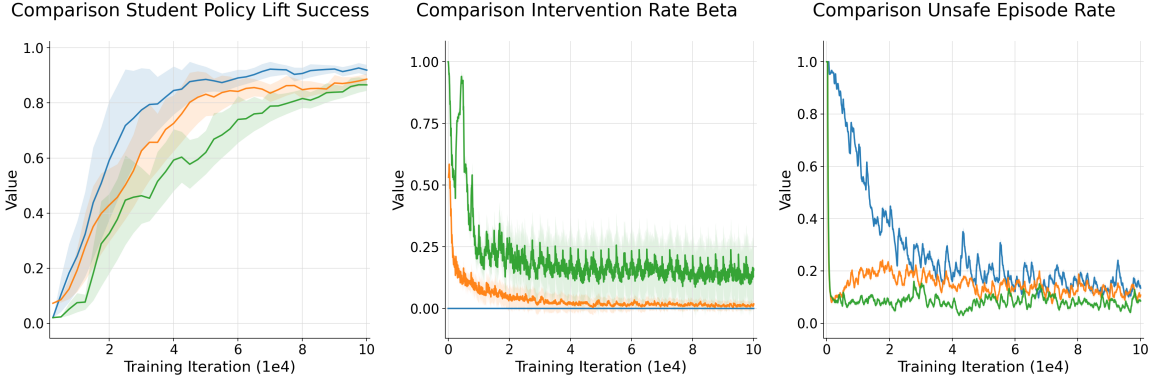


Fig. 3: **Ablation studies.** We compare three distillation variants in terms of (left) student lift success, (middle) teacher intervention rate β , and (right) unsafe episode rate, all plotted over training iterations. Shaded regions indicate variability across runs (mean $\pm$ std). Curves correspond to the three compared methods (blue: vanilla DAGger, yellow: vanilla SafeDAGger, green: our pipeline).

($\beta = 0$), while the teacher runs in parallel to provide labels: each executed student action a_t^S is relabeled with the corresponding teacher action a_t^T .

(b) Our RL teacher + Vanilla DAGger. This setting matches (a) except that the teacher is trained with our customized RL recipe instead of the DexTraH-RGB pipeline.

(c) Our RL teacher + Vanilla SafeDAGger. Using the same teacher as in (b), we replace Vanilla DAGger with a disagreement-gated SafeDAGger baseline. The teacher intervenes only when $\|a_t^T - a_t^S\|_2$ exceeds a threshold, and we train the student on teacher-labeled pairs (o_t^S, a_t^T).

(d) DexTraH-RGB RL teacher + SafeDAGger with disagreement and a risk predictor. This setting matches our full distillation pipeline but uses the DexTraH-RGB-trained teacher (as in (a)). Intervention is triggered by either large teacher-student disagreement or a high predicted risk of failure, and the student is trained from teacher-labeled pairs (o_t^S, a_t^T) collected under this gate.

(e) Our RL teacher + SafeDAGger with disagreement and a risk predictor. This is our full pipeline: the teacher is trained with our customized RL recipe (as in (b)), and distillation uses the same intervention gate as in (d), combining teacher-student disagreement and state-risk prediction to trigger teacher takeover.

As shown in Table IV, ablations (a) and (d), which use the DexTraH-RGB RL recipe, exhibit lower lift success and higher unsafe rate under both Vanilla DAGger and our intervention mechanism. We attribute this gap to limited transferability of the DexTraH-RGB teacher training setup: it relies heavily on a FABRICS controller and was tuned for the KUKA-Allegro platform. When applied to our hardware and task setting, the resulting teacher policies degrade due to mismatched dynamics and nontrivial hyperparameter retuning. Consequently, the weaker teacher policy set constrains student distillation performance in downstream training.

As shown in Fig. 3, ablations (b), (c), and (e), which adopt our RL pipeline, reach a comparable final lift success on the

TABLE IV: Ablation performance across various objects (tested over 200 episodes). Each ablation reports lift success and average unsafe rate.

Percentage %	WindCart		Boot		ToolBox		Cow	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
Ablation (a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5
Ablation (b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5
Ablation (c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0
Ablation (d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0
Ablation (e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0

training objects (84%–89%; left). This is expected because our intervention mechanism is designed primarily to suppress unsafe exploration rather than directly optimize task success. Vanilla DAGger keeps the student in full control ($\beta = 0$), whereas both SafeDAGger variants exhibit a decreasing intervention rate as the student improves (middle). Both intervention-based methods substantially reduce unsafe episodes (by up to 80% compared to Vanilla DAGger) early in training and maintain a low unsafe rate (below 20%) throughout distillation, with the proposed method achieving the lowest unsafe rate overall (right).

Compared with Vanilla SafeDAGger, our method intervenes more frequently because the risk predictor expands the set of transitions deemed unreliable, leading to additional teacher takeovers. This increased supervision further reduces the unsafe rate (by about 5% on average). By the end of training, our method converges toward the teacher policy’s average unsafe rate ($\approx 12\%$) and reaches a minimum unsafe rate ($\approx 8\%$) (both averaged over the evaluation objects), which is below the teacher average.

As shown in Fig. 4, the per-object failure-mode decomposition indicates that unsafe episodes are dominated by harmful collisions and object escape across most objects. Relative to vanilla DAGger, both intervention-based methods substantially reduce the per-object unsafe rate and consistently suppress these two dominant failure modes. Moreover, disagreement-

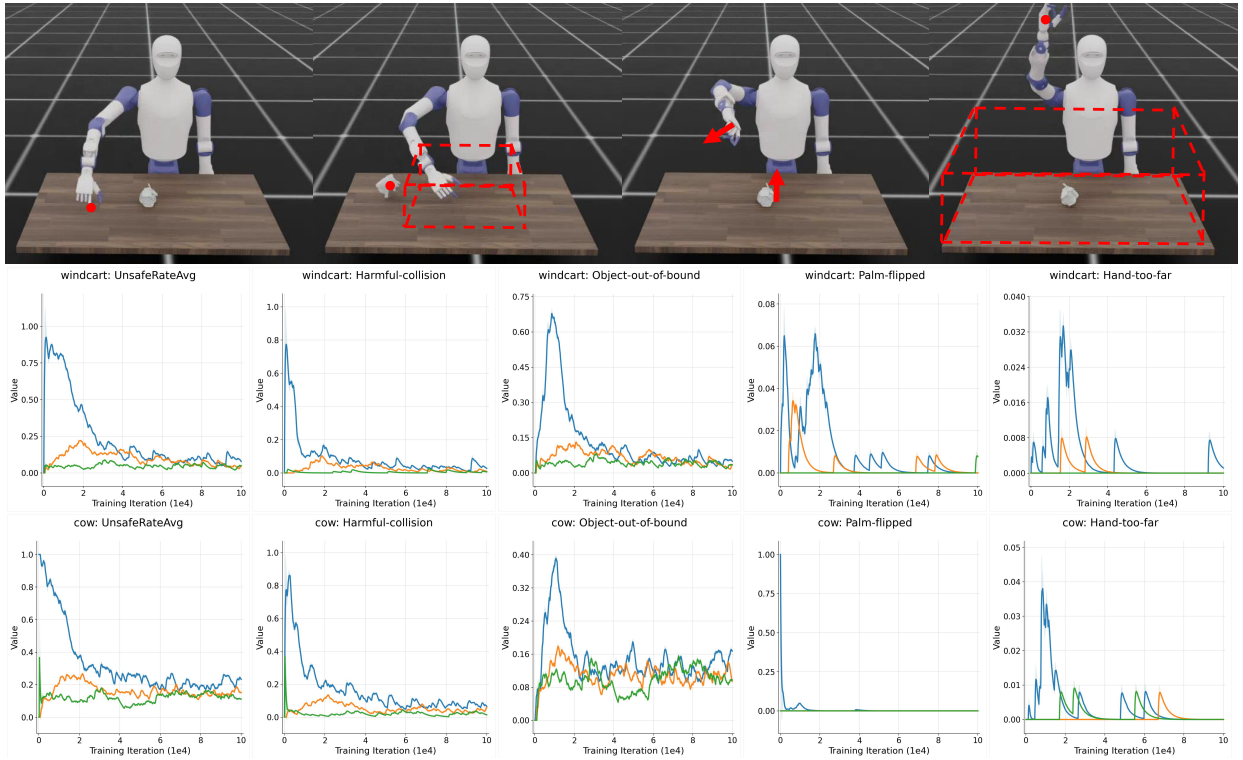


Fig. 4: **Per-object failure-mode decomposition during online distillation ablations.** We illustrate four task-specific failure modes in simulation (harmful collision, object escape, palm flipped, and hand too far) and report the aggregate unsafe rate together with its per-mode breakdown over training iterations. Results are shown for two objects (windcart and cow); lower is better in all plots. Curves correspond to the three compared distillation variants (blue: vanilla DAgger[10], yellow: vanilla SafeDagger[12], green: our pipeline).

based SafeDagger already eliminates many of the large early spikes in collision and escape, suggesting that teacher takeovers effectively prevent the student from entering highly unsafe regions of the state space during the initial distillation phase.

The combination of teacher-student disagreement with the risk predictor further improves safety, yielding the lowest unsafe rate for most objects. This gain is primarily attributable to additional reductions in collision- and escape-induced terminations, especially in cases where disagreement alone is insufficient to anticipate downstream failures. Overall, the decomposition supports our design choice: augmenting disagreement with a learned risk signal provides a stricter intervention criterion that reduces unsafe exploration while maintaining competitive task performance.

D. Real-World Experiments Setup

The experiments exhibit a lower lift success rate and a higher unsafe episode rate than the teacher and student evaluation results in simulation, indicating that the proposed pipeline does not fully eliminate the sim-to-real gap. Nevertheless, the resulting policy is sufficiently robust for real-world deployment.

The unseen-object evaluation reveals a clear generalization gap: compared to seen objects (WindCart/Boot/ToolBox/Cow), the policy achieves lower lift success and higher unsafe rates on unseen objects (Plane/Shoe). This degradation is primarily driven by increased harmful collisions (20.0% and 26.7% on

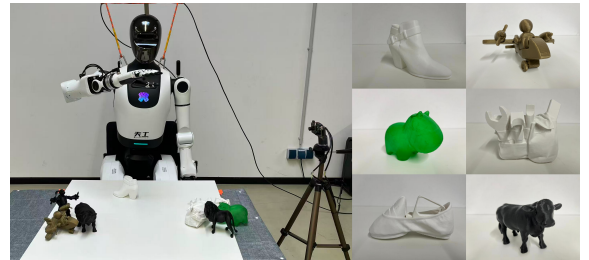


Fig. 5: **Real-world experimental setup and test objects.** Left: the single-arm manipulation platform used for deployment and evaluation. Right: examples of the physical objects used in the real-world lift experiments.

Plane and Shoe), while object escape remains comparable across objects (≈ 6.7 – 13.3%). No palm-flip or hand-too-far events are observed, suggesting that real-world failures are dominated by contact-rich interactions rather than gross pose instabilities. Overall, the results indicate that the remaining sim-to-real gap on unseen geometries is largely attributable to a contact-strategy mismatch, i.e., the policy fails to adapt its contact interactions to novel object shapes.

E. Discussion

While effectively reducing unsafe exploration, the teacher-intervened online distillation process appears to reduce sample efficiency. In Fig. 3, we observe slower performance growth in lift-success evaluation under higher intervention rates. The

TABLE V: Real-world experiments across various objects (tested over 15 episodes). **Unseen objects** are shaded.

Percentage %	WindCart	Boot	ToolBox	Cow	Plane	Shoe
Lift Success	86.7	86.7	80.0	86.7	66.7	60.0
Unsafe Rate Avg	13.3	20.0	20.0	26.7	40.0	40.0
Harmful collision	0.0	6.7	13.3	13.3	20.0	26.7
Object escape	13.3	13.3	6.7	13.3	20.0	13.3
Palm flipped	0.0	0.0	0.0	0.0	0.0	0.0
Hand too far	0.0	0.0	0.0	0.0	0.0	0.0

underlying cause remains unclear; we believe there is room for improvement in both exploration safety and sample efficiency within the online distillation paradigm.

V. CONCLUSION AND FUTURE WORK

In this work, we propose a combined intervention mechanism that integrates teacher-student disagreement with a learned risk predictor. The proposed method effectively improves exploration safety in a multi-teacher distillation process. Through extensive evaluation, we validate that the improvement is robust across all test objects and all failure modes that we consider important. Our method also appears to approach the teacher policy’s performance ceiling in avoiding unsafe regions. Successful real-world deployment further shows that the resulting student policy partially bridges the sim-to-real gap; however, poor generalization to unseen objects indicates that our pipeline is vulnerable to unvisited, complex contact scenarios.

A clear limitation is that performance during distillation, as well as the final student policy, is largely constrained by teacher capability. Developing a safety-aware teacher training pipeline is a promising direction for future work to generate stronger teacher policies.

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